

Figure 1: The traditional IT environment

Standards and Technology), and verify if the private cloud measures up on the five essential characteristics listed by it.

- **On-demand self-service:** A consumer can provision computing capabilities such as compute resources, the network and storage as needed, without human interaction in the private cloud environment.
- **Broad network access:** Resources are accessible over the organisational network, behind a firewall, and accessible through standard mechanisms that promote use by heterogeneous thin or thick client platforms such as mobile phones, tablets, laptops and workstations.
- **Resource pooling:** In private clouds, computing resources are pooled to serve the multiple business units of the same organisation.
- **Rapid elasticity:** Resource capabilities can be elastically provisioned and de-provisioned, in some cases automatically, to scale up and scale down based on the demand. To the consumer, the capabilities available for provisioning often appear to be unlimited and can be appropriated in any quantity at any time. Though, in the case of private clouds, capabilities available for provisioning may not appear to be unlimited, considering the huge investment required, which might be beyond a single organisation's budget.
- **Measured service:** Compute, storage and network resource usage can be monitored extensively, controlled and reported, providing transparency for both the organisation and its various business units.

Private cloud platform vendors with virtualisation, grid and infrastructure solutions backgrounds have focused on adding features like multi-tenancy, self-service and charge-back. We see these capabilities enabling organisations to go well beyond consolidation and virtualisation projects to set up their own clouds.

"The growth of the private cloud is being driven by pressure to deliver IT faster and cheaper, the quick penetration of virtualisation and virtualisation management, and the growth of cloud computing offerings," said Tom Bittman, VP and analyst at Gartner. Server and infrastructure virtualisation are important basics for the private cloud. Virtualisation



Figure 2: The private cloud environment

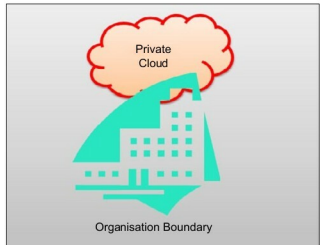


Figure 3: The private cloud

makes it easier to dynamically and granularly pool, share and reallocate infrastructure resources such as servers, desktops, storage, networking, middleware, etc. Hence, virtualisation is the major entry point for the private cloud. However, the trap that executives often fall into is when they believe virtualisation is a cloud environment. By itself, it does not constitute a private cloud.

The private cloud is a blend of automation, virtualisation and distributed computing that provides on-demand computing capacity to an organisation's internal users or customers. In a cloud environment, security is the single most fundamental consideration. Dynamic provisioning, confidentiality, governance, risk management and compliance are the key driving factors for the private cloud. Private clouds are slowly moving from a technology-centric to a user-centric service delivery. Most private cloud initiatives have so far been technology-centric, focused on enabling automation, virtualisation and service oriented architecture (SOA), but now the focus is on standardisation, automation and to make organisations financially aware.

Application and data that are migrated to a private cloud would deliver flexible computing similar to any other cloud service provider, but behind a firewall and